Assignment #1: Quarto Data Analysis

Assigned Thursday 16/04/2026 Due Monday 20/04/2026

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Work with a partner!

1. Download and read in *p301.csv* - find it under materials for Week 1
2. In an Quarto document, complete two data-analysis tasks (whatever you think might be interesting)
3. In your document, describe your motivation, and comment on results
4. Render your Quarto complete with code, comments and results
5. Submit *pdf* on Canvas for Assignment 1

You can be creative here - you could summarize precipitation by water year, find the mean annual precipitation and then graph annual time series (don't pick this one, since I suggested it :) You could look at the relationship between two variables using regression or correlation; you could split the data in some way and do a t-test to look at whether means are different; you could look at trends. Its up to you

The data

p301.csv

Climate and estimates of water and forest carbon information from USFS experimental watershed in the Southern California Sierra (above Fresno)

Variable	Description	Units

What the variables are (metadata)

Variable	Description	Units
month	Month	
year	Year	
day	Day	
wy	Water Year (Oct–Sept hydrologic year)	
tmax	Maximum daily temperature	°C
tmin	Minimum daily temperature	°C
precip	Depth of precipitation	mm
transpiration	Estimated forest transpiration	$m/m^2/day$
evaporation	Estimated evaporation	$m/m^2/day$
NPP	Estimated net carbon uptake by trees	$kgC/m^2/day$
plant_carbon	Estimated standing biomass in units of carbon	kg/m^2
streamflow	Streamflow (per watershed area)	$m/m^2/day$
snowpack	Amount of snow on the ground (water equivalent)	m/m^2
direct solar	Incoming direct solar radiation	$kJ/m^2/day$
diffuse solar	Incoming diffuse solar radiation	$kJ/m^2/day$